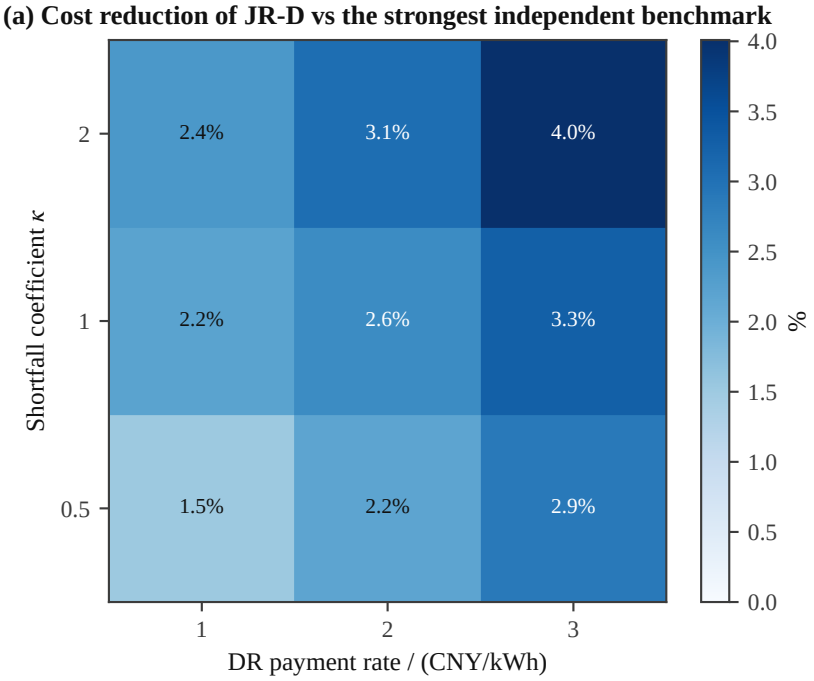
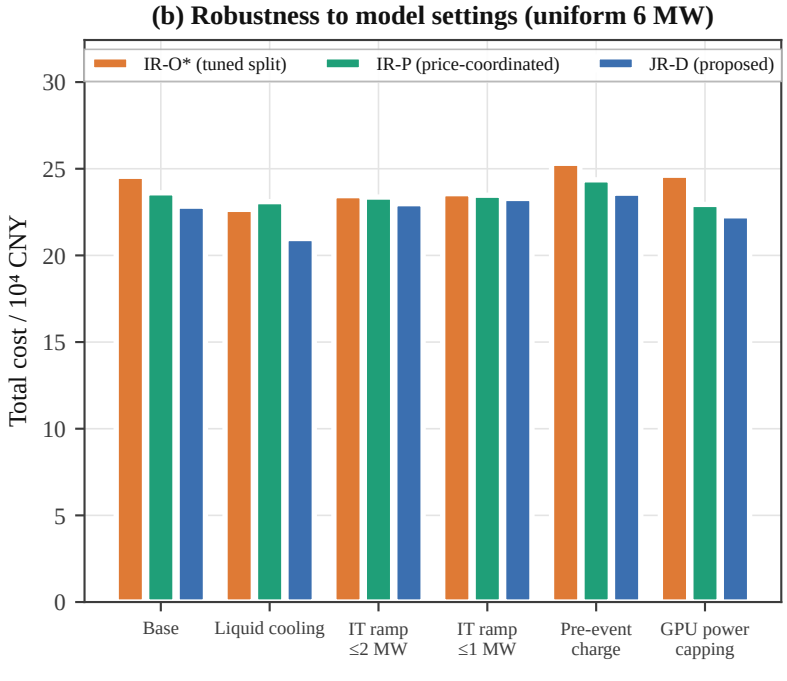

(a) Cost reduction of JR-D vs the strongest independent benchmark
2.4%
3.1%
4.0%
2.2%
2.6%
3.3%
1.5%
2.2%
2.9%
Shortfall coefficient κ
2
1
0.5
1
2
3
DR payment rate / (CNY/kWh)
%
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0


(b) Robustness to model settings (uniform 6 MW)
IR-O* (tuned split)
IR-P (price-coordinated)
JR-D (proposed)
Total cost / 10⁴ CNY
30
25
20
15
10
5
0
Base
Liquid cooling
IT ramp ≤2 MW
IT ramp ≤1 MW
Pre-event charge
GPU power capping